“Nazarbayev Intellectual School of Science and Mathematics in Taldykorgan”, branch of Autonomous Educational Organization “Nazarbayev Intellectual Schools”.

Authors: Tolegen Saltanat, Grade 11
Oralbek Shakhnur, Grade 11

Uranium Recovery

Supervisor: Avdyunin Nikolay Vladimirovich
Head of the Extended Laboratory of Engineering and Technical Creativity and Information Technologies

Taldykorgan, 2026

Table of contents:

1.Abstract.....3

2.Introduction.....5

3.Literature Review. Theoretical Section.....6

3.1 Uranium and Its Significance.....6

3.2 Main Methods of Uranium Extraction.....6

3.3 Major Challenges and Environmental Issues in Uranium Mining.....6

4. Experimental Section.....7

4.1 Research Procedure, Reagents, and Equipment.....7

4.2 Conducting Experiments Using the New Method.....7

4.3 Development of an Autonomous Robot Model.....8

4.4 Comparison with Traditional Methods.....8

4.5 Economic and Environmental Analysis.....9

5. Conclusion.....10

6. References.....11

1.Abstract

This scientific paper considers the issue of obtaining uranium from seawater in an environmentally safe and effective way. The work was carried out independently: a research plan was drawn up, literature data was analyzed, copper ions were used as an experimental model, and a new approach was proposed with an autonomous robot model combining adsorption and electrochemical desorption methods.

The purpose of the scientific project is to propose a model of a robot operating in an autonomous system by studying the main methods of extracting uranium from seawater, combining the most effective ones and carrying out improvement work.

Hypothesis: The combination of adsorption and electrochemical methods increases the efficiency of uranium extraction and ensures environmental safety (reagent-free regeneration).

Research progress:

1) analysis of the theoretical foundations and environmental aspects of uranium mining by collecting literature data;

2) experiment in the laboratory using the main selected methods of electrochemistry and adsorption;

3) Representation of an autonomous robot model;

4) experimental data processing, comparison and summing up;

Experimental methodology:

1) Adsorption of Cu^{2+} ions by natural sorbents;

2) separation of ions from solution by electrochemical method;

3) Evaluation of the effectiveness of combining methods;

The novelty of the research:

In the study, adsorption and electrochemical regeneration were combined into a single system, and the sorbent became part of the electrode, serving as a filter. Instead of a stationary installation, a model of an autonomous robot was developed that independently took and processed water samples. Instead of using solid acids, electrochemical desorption was used.

Main results: Cu^{2+} ions were captured by a natural sorbent with an efficiency of 70–85%; the electrochemical method allowed the sorbent to be regenerated without acid; the process needs no heating or acid treatment, so it is expected to use less energy than traditional approaches. The principle of operation of an autonomous robot is also simulated.

Practical significance: the method serves as the basis for the development of environmentally safe and sustainable technologies for the extraction of uranium from seawater. The robotic system can also be used to remove heavy metals and purify water resources.

2.Introduction

Despite the rapid development of “green energy” in the 21st century, nuclear energy remains one of the most reliable and powerful sources of electricity generation. Currently, the world’s oceans contain approximately 4.5 billion tons of uranium, which is nearly 1000 times greater than terrestrial reserves [1]. According to the latest OECD NEA/IAEA “Red Book” data, identified land-based uranium resources amount to approximately 5.9 million tons, while the world’s reactors need nearly 69 thousand tons of uranium every year [2]. These figures indicate that terrestrial resources alone cannot meet long-term global demand.

At the national level, Kazakhstan is a global leader in uranium production, accounting for about 40% of global output and holding about 14% of the world’s uranium resources [3]. However, conventional uranium mining methods cause significant environmental damage, including soil degradation, groundwater and surface water contamination, and accumulation of radioactive waste. Therefore, the development of environmentally safe and alternative uranium extraction methods is highly relevant both for Kazakhstan and globally.

The concept of uranium extraction from seawater has attracted significant scientific interest. Although the concentration of uranium in seawater is low (approximately 3.3 μg/L), the vast volume of the oceans makes this resource virtually inexhaustible [1]. However, existing methods such as ion-exchange, membrane technologies, and conventional sorption either demonstrate low efficiency or require high costs and aggressive chemical reagents. Consequently, there is a need to develop a new approach that integrates the advantages of different methods while minimizing their drawbacks.

Research objective:

To propose an autonomous robotic system that integrates adsorption and electrochemical desorption/regeneration methods for environmentally safe and energy-efficient extraction of uranium ions from seawater.

Research tasks:

To analyze existing uranium extraction methods, assess the potential of adsorption and electrochemical techniques, develop an autonomous robotic model for water sampling and processing, and compare the efficiency of the proposed method with traditional approaches.

Research methods:

1. Analysis of scientific literature and statistical data;
2. Laboratory experiments on adsorption and electrochemical desorption of Cu^{2+} ions;
3. Modeling of an autonomous robotic system for marine applications.

3.Literature Review. Theoretical Section

3.1 Uranium and Its Significance

Uranium is the primary raw material for nuclear energy production. It provides extremely high energy density: one kilogram of natural uranium used in a reactor releases roughly as much energy as burning about 20 tons of coal. Uranium is also used in space technologies, medicine, and the defense industry.

The world's oceans are estimated to contain about 4.5 billion tons of uranium [1], which is nearly 1000 times greater than confirmed terrestrial reserves. Identified land-based resources amount to approximately 5.9 million tons, while annual world demand is nearly 69 thousand tons [2]. Therefore, uranium extraction from seawater represents a strategic solution for future energy security. Kazakhstan alone accounts for about 40% of global uranium production [3].

3.2 Main Methods of Uranium Extraction

1) **Conventional methods:** Ion-exchange resins and membrane technologies are widely used but require large amounts of reagents, generate radioactive waste, and pose environmental risks [1].
2) **Adsorption:** This method utilizes porous sorbents, such as amidoxime-functionalized fibers or natural minerals, to capture uranium ions [1]. Its advantages include simplicity, while drawbacks include slow kinetics and frequent sorbent regeneration.
3) **Electrochemical methods:** By applying voltage to electrodes, uranyl ions (UO_2^{2+}) are reduced at the cathode and deposited in metallic form. This approach avoids the use of acids, reducing waste generation [4], but requires relatively high energy input.
4) **Metal-organic frameworks (MOFs):** Nanoporous materials such as UiO-68-type frameworks show high uptake of uranium from water [5]. However, their high synthesis cost and long-term instability limit practical application.
5) **Biosorption:** Microorganisms and biopolymers such as chitosan can absorb uranium ions. While environmentally friendly, this method faces limitations in stability and large-scale industrial use [6].

3.3 Major Challenges and Environmental Issues in Uranium Mining

The extremely low concentration of uranium in seawater (approximately 3.3 μg/L) [1] requires processing large volumes of water. Existing methods are costly, slow, unstable, and often generate waste. A major environmental concern is the risk of chemical reagents entering marine ecosystems.

4. Experimental Section

4.1 Research Procedure, Reagents, and Equipment

Due to safety concerns related to uranium handling, copper ions were used as a model substance.

- Solution 1: Alakol Lake water
- Solution 2: Dilute $CuSO_4$ solution
- DC power supply
- Graphite anode and cathode
- Adsorbent based on activated carbon and chitosan

4.2 Conducting Experiments Using the New Method

1) Graphite electrodes were installed and connected to a DC power supply.
2) Two solutions of equal volume ($CuSO_4$ solution and Alakol Lake water) were prepared.
3) The adsorbent was added to both solutions.
4) Electrolysis was conducted while observing color and precipitate changes.
5) After 10–15 minutes, a light pink layer of metallic copper (reduced Cu^{2+} ions) appeared on the cathode surface.

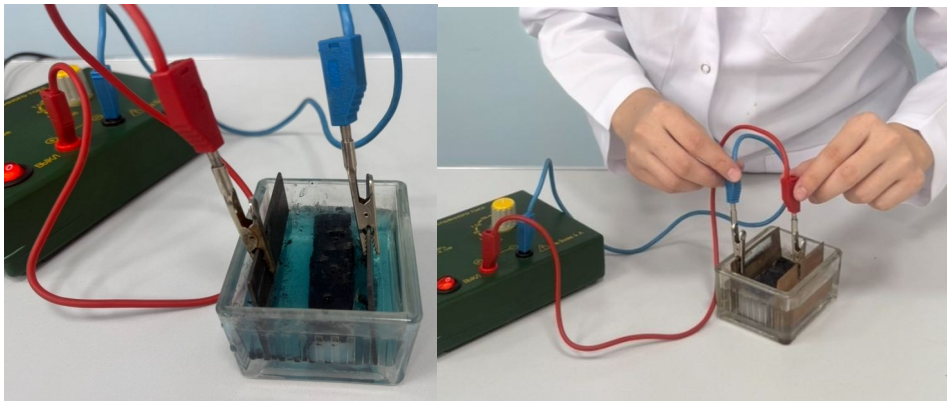


Figure 1. Experimental setup: graphite electrodes and the sorbent in the solution, connected to a DC power supply.

Observations:

- In the $CuSO_4$ solution, the reaction proceeded faster, with copper depositing on the cathode and the solution becoming lighter.
- In Alakol water, the process was slower, but copper particles eventually appeared on the sorbent surface.

- Darkening of the sorbent indicated adsorption of metal ions.

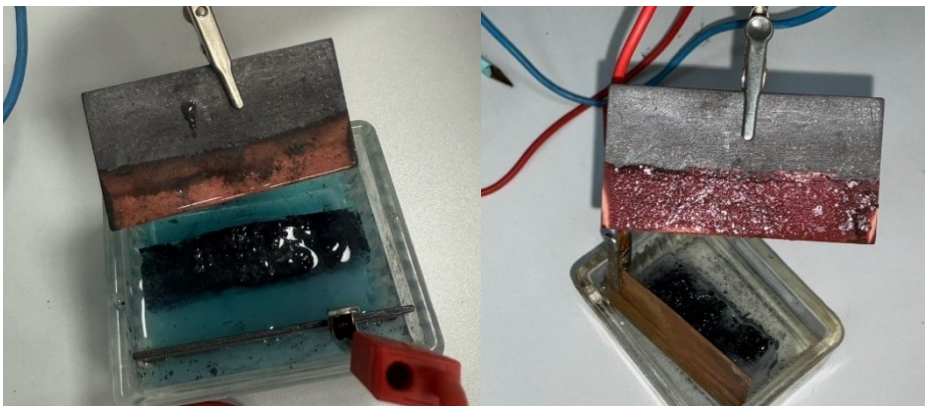


Figure 2. After electrolysis: a pink layer of copper on the cathode and the darkened sorbent.

At the cathode (reduction):

$Cu^{2+} + 2e^- \rightarrow Cu(s)$

At the anode (graphite):

$2H_2O \rightarrow O_2(g) + 4H^+ + 4e^-$

Overall reaction:

$Cu^{2+} + H_2O \rightarrow Cu(s) + ½O_2(g) + 2H^+$

Conclusion of laboratory experiments:

The combined adsorption–electrolysis system effectively extracts metal ions even from natural waters with low metal concentrations, confirming the applicability of the proposed method for rare and heavy metal recovery.

4.3 Development of an Autonomous Robot Model

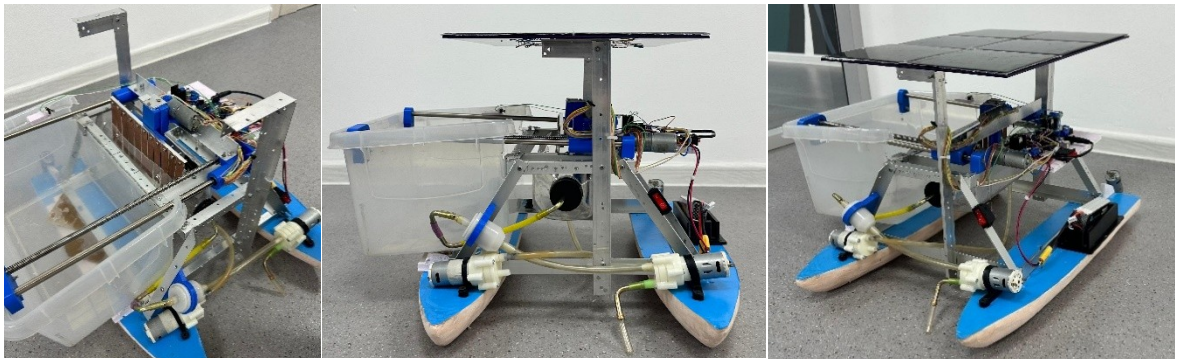


Figure 3. The autonomous robot model: a catamaran with a solar panel, water pumps and the adsorption–electrochemical module.

The proposed robot autonomously collects water samples and processes them using an adsorption–electrochemical module. The extracted concentrate is stored in a sealed container. The system is equipped with remote control and monitoring technologies.

4.4 Comparison with Traditional Methods

Traditional Methods	**Proposed Method**
Require large amounts of reagents, generate waste, use hazardous acids and solvents, consume high energy, and are costly and time-consuming	Based on natural sorbents, low-cost and accessible, reagent-free, acid-free, fast reaction rate, low energy consumption, and waste-free

Experimental results showed that 70–85% of Cu^{2+} ions were extracted and regenerated, demonstrating competitiveness with traditional adsorption methods.

4.5 Economic and Environmental Analysis

The proposed method is economically efficient, as it does not require expensive reagents. Costs are primarily associated with electricity and sorbent materials. The autonomous robotic system reduces labor requirements, lowering operational costs. High process speed further reduces the overall cost of uranium extraction and facilitates long-term industrial scalability.

From an environmental perspective, the method is safe: electrochemical desorption replaces strong acids, natural sorbents minimize environmental impact, and the closed robotic system prevents the release of radioactive or toxic substances. Lower expected energy consumption could also reduce indirect carbon emissions, contributing to sustainable uranium extraction from seawater.

5. Conclusion

This study proposes a novel approach to uranium extraction from seawater based on the integration of adsorption and electrochemical methods. Due to safety considerations, Cu^{2+} ions were used as a model substance.

Experimental results demonstrated that adsorption effectively captures metal ions, while electrochemical regeneration enables acid-free reuse of sorbents. The combination of these methods provides a resource-efficient and environmentally safe system.

The proposed autonomous robot can transmit data remotely via a mobile application and independently collect and process water samples. This reduces reliance on stationary installations used in countries such as the USA and China, enabling application in marine and remote environments.

Practical significance of the project:

- Contributes to the development of environmentally safe uranium extraction technologies from seawater;
- Provides a foundation for improving technologies combining sorption and electrochemical regeneration;
- Enables application of autonomous robotic systems in water purification and heavy metal removal.

6. References

[1] Abney C.W., Mayes R.T., Saito T., Dai S. Materials for the recovery of uranium from seawater // Chemical Reviews. – 2017. – Vol. 117, №23. – P. 13935–14013.

[2] World Nuclear Association. Supply of Uranium [Electronic resource]. – URL: https://world-nuclear.org/information-library/nuclear-fuel-cycle/uranium-resources/supply-of-uranium (accessed: 31.10.2026).

[3] World Nuclear Association. Uranium and Nuclear Power in Kazakhstan [Electronic resource]. – URL: https://world-nuclear.org/information-library/country-profiles/countries-g-n/kazakhstan (accessed: 31.10.2026).

[4] Liu C., Hsu P.-C., Xie J., Zhao J., Wu T., Wang H., Liu W., Zhang J., Chu S., Cui Y. A half-wave rectified alternating current electrochemical method for uranium extraction from seawater // Nature Energy. – 2017. – Vol. 2, №4. – Art. 17007.

[5] Carboni M., Abney C.W., Liu S., Lin W. Highly porous and stable metal–organic frameworks for uranium extraction // Chemical Science. – 2013. – Vol. 4, №6. – P. 2396–2402.

[6] Wang J., Chen C. Chitosan-based biosorbents: modification and application for biosorption of heavy metals and radionuclides // Bioresource Technology. – 2014. – Vol. 160. – P. 129–141.